

Homework: Risk Analysis Report IT Project Management

Exercise 1: EMV and Decision Tree Analysis

Student Name: Trần Anh Tuấn

Student ID: 24521943

Course ID: MSIS3033.Q21.CTTT

April 21, 2026

1 EMV Calculation (Expected Monetary Value)

The EMV is a statistical concept that calculates the average outcome when the future includes uncertain scenarios.

$$EMV = \sum (\text{Probability} \times \text{Net Profit})$$

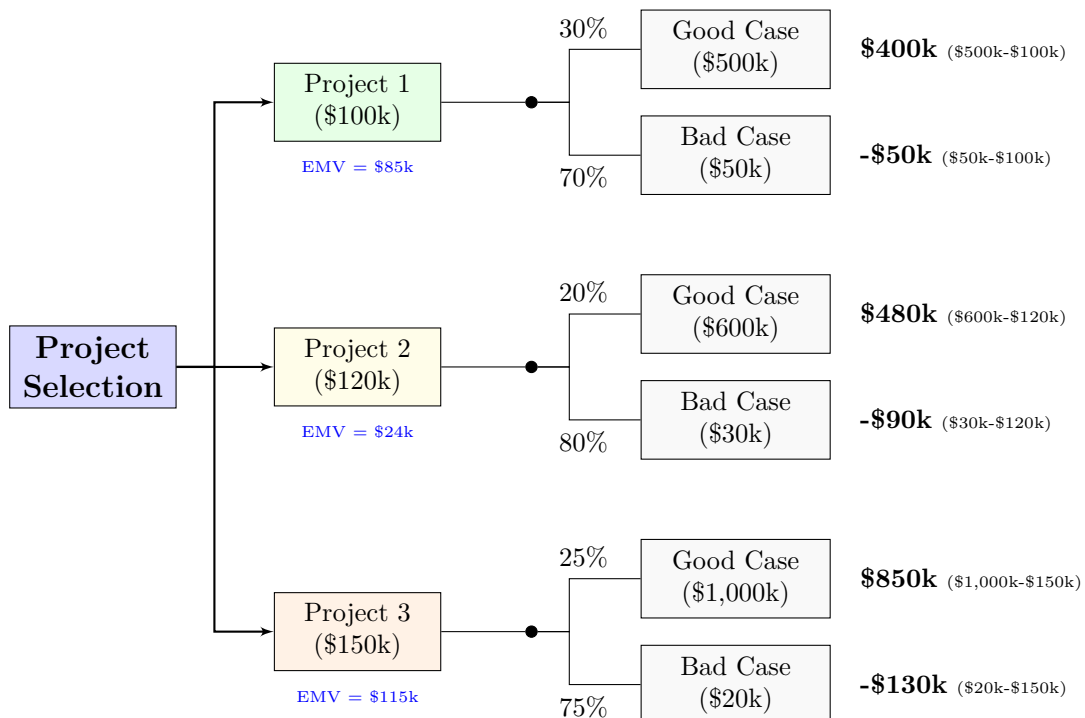
Note: Net Profit = Revenue - Investment Cost.

1.1 Project Profitability Breakdown

Project	Scenario	Net Profit (V)	Probability (P)	P × V
Project 1 (Cost: \$100k)	Good Case	\$400,000	30%	\$120,000
	Bad Case	−\$50,000	70%	−\$35,000
Total EMV Project 1				\$85,000
Project 2 (Cost: \$120k)	Good Case	\$480,000	20%	\$96,000
	Bad Case	−\$90,000	80%	−\$72,000
Total EMV Project 2				\$24,000
Project 3 (Cost: \$150k)	Good Case	\$850,000	25%	\$212,500
	Bad Case	−\$130,000	75%	−\$97,500
Total EMV Project 3				\$115,000

2 Decision Tree Analysis

The following diagram illustrates the decision paths, chance nodes, and final profit outcomes for each project.



3 Conclusion

- **Best Financial Return: Project 3** has the highest EMV (\$115,000). While it requires the largest investment, its "Good Case" payoff is significant enough to outweigh the potential losses on average.
- **Lowest Risk Choice: Project 1** is the most conservative and lowest-risk option.
 - It has the lowest potential "Max Loss" in a bad scenario (-\$50,000).
 - It has the highest probability of success (30% chance for the good case).